



Predictive Modeling of Root-Zone Variables Using Greenhouse Data

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01 Motivation

Root-zone pH and electrical conductivity (EC) indicate nutrient status, salinity, and fertilizer effects. In greenhouses, they are often measured manually and infrequently, leaving gaps where irrigation, fertigation, climate, and plant uptake changes may be missed. Predictive modeling can estimate pH and EC under planned irrigation/fertigation actions, helping growers adjust management before conditions drift.

02 Project Goal

- Develop a soft sensor that predicts future root-zone pH and EC using the latest pH/EC measurement, forecast climate, planned irrigation/fertigation, and crop-state data.
- Estimate $t_0 \rightarrow t_1$ changes in pH and EC, allowing growers to evaluate planned irrigation/fertigation strategies and adjust management before conditions drift.

03 Data

16,682

10-min records

109

pH/EC samples

20+

input features

Unified 10-minute master dataset, May 29 – Sep 21, 2025:

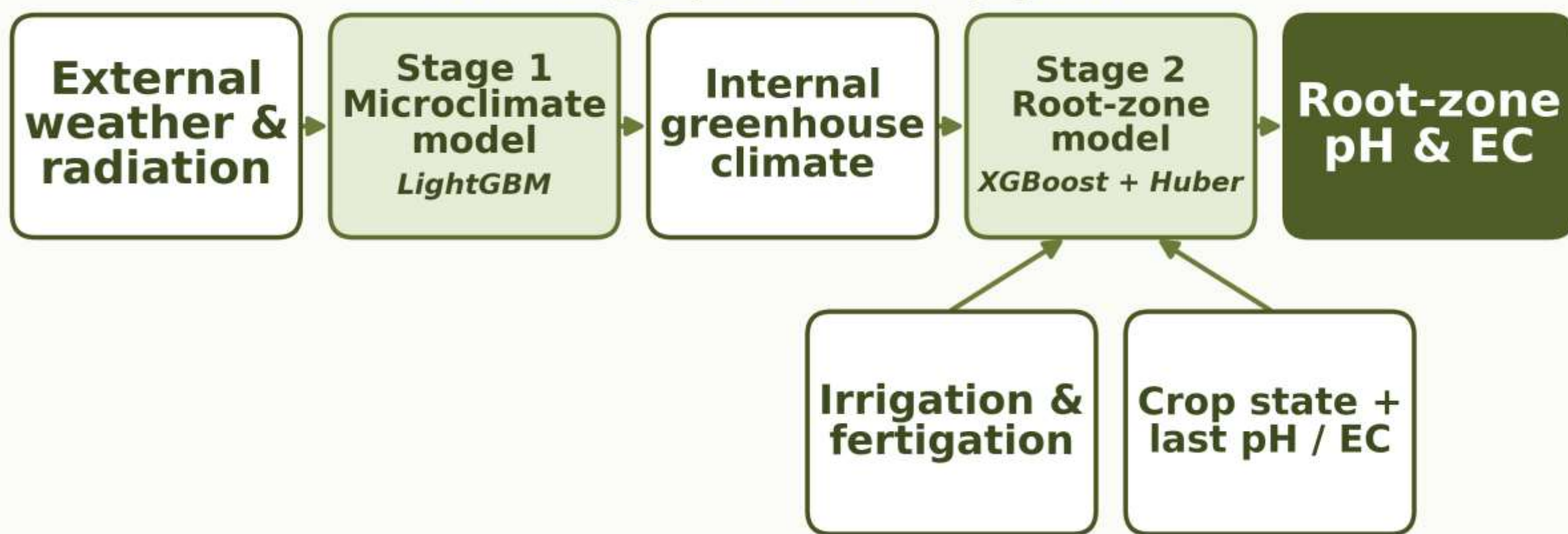
- External weather & solar radiation — weather station and solar sensor
- Greenhouse microclimate — internal sensor data
- Irrigation & fertigation — scheduled management inputs
- Crop-state features — canopy-cover observations
- Manual pH & EC measurements — lab-measured model labels

04 Method

Two-stage prediction pipeline:

- Forecast internal greenhouse microclimate from external weather and radiation.
- Predict $t_0 \rightarrow t_1$ Δ pH/ Δ EC from forecast climate, crop state, and planned events.

Two-stage prediction pipeline



05 Algorithms & Model Choice

Hybrid model choice: nonlinear prediction + robustness to sharp-change intervals.

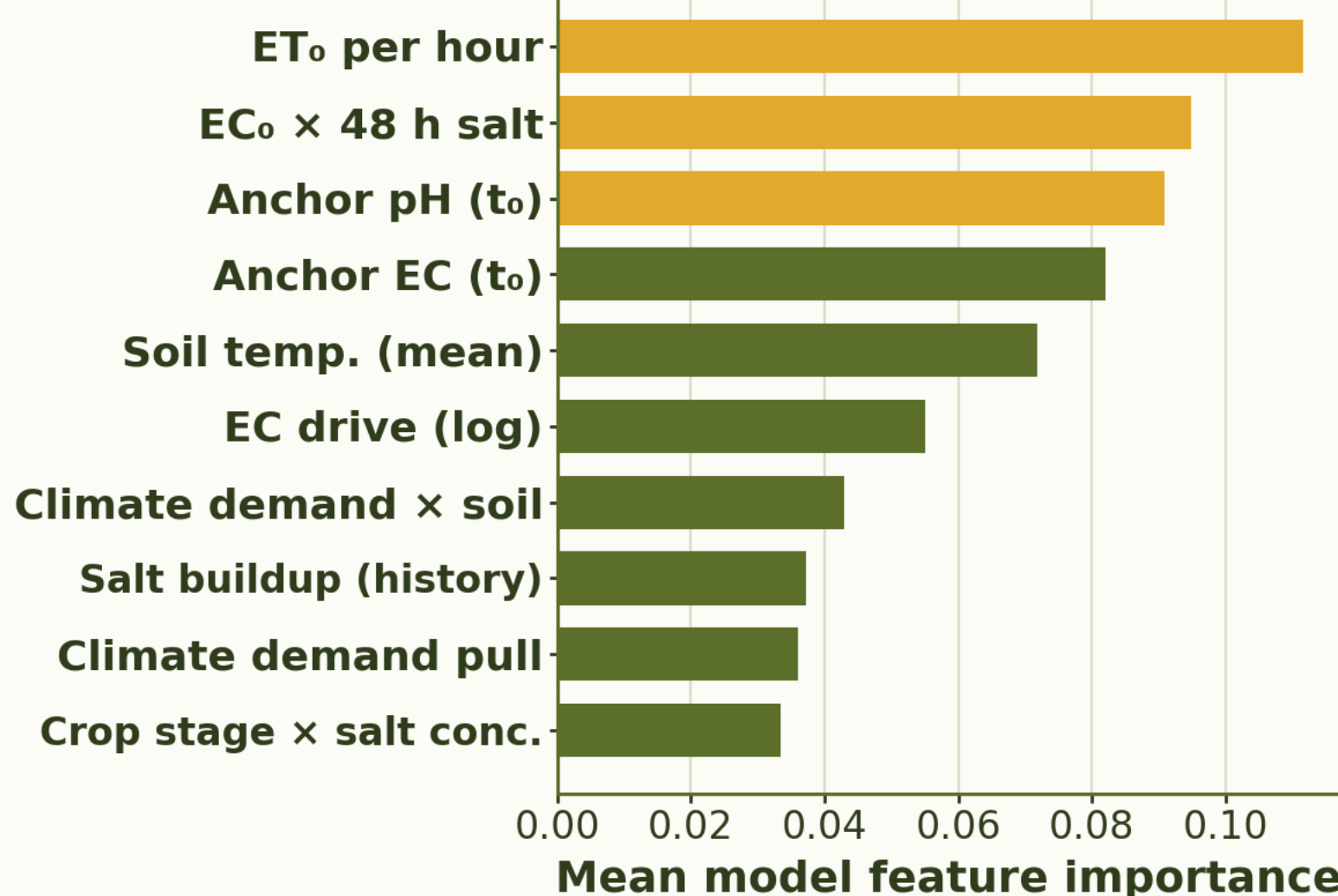
Gradient Boosting

Main nonlinear model for tabular greenhouse data. Captures interactions between climate, irrigation, fertigation, crop state, and baseline pH/EC.

Huber Regressor

Robust companion model for sharp pH/EC shifts. Reduces sensitivity to extreme intervals linked to acid, fertigation, or salinity changes.

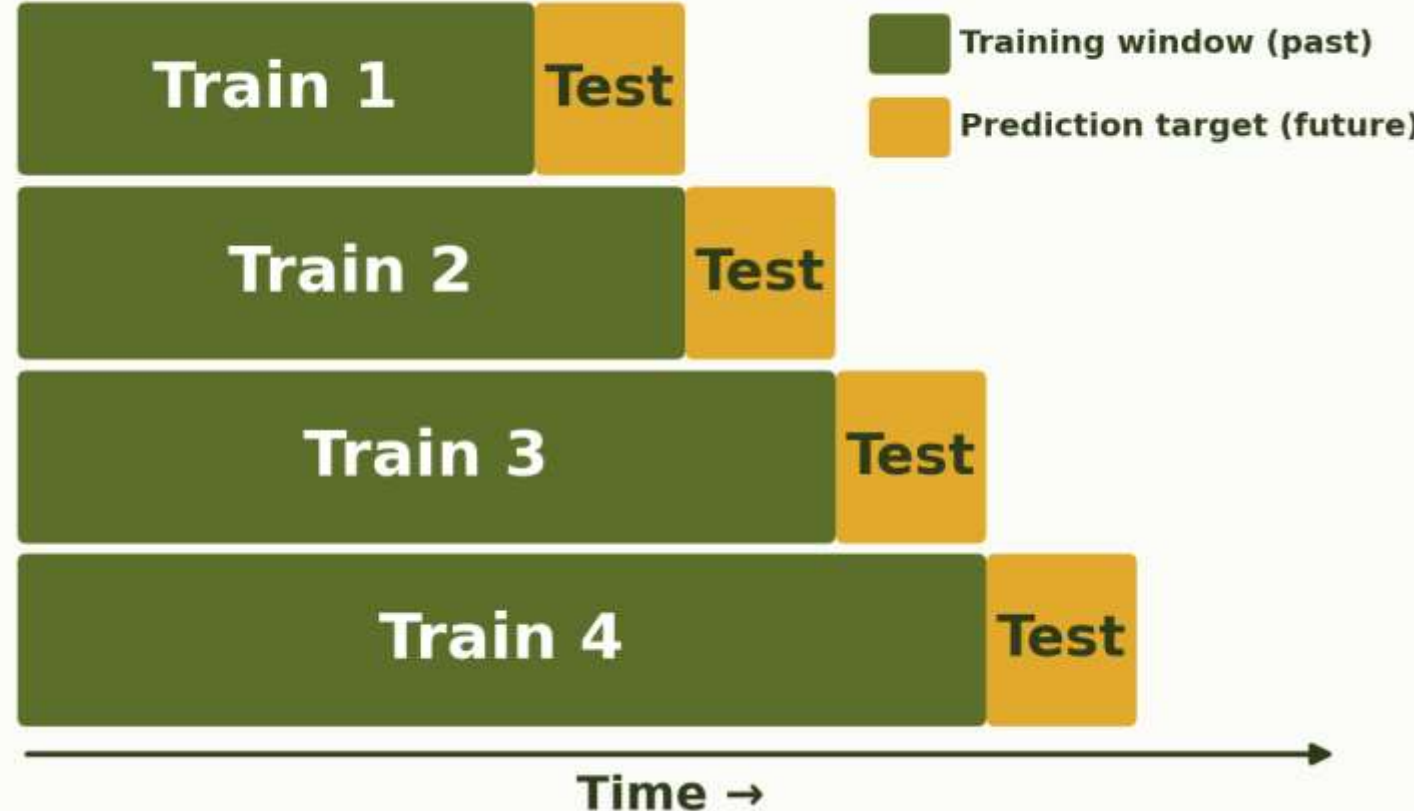
Most informative features



06 Validation Strategy

- Walk-forward validation matches deployment: future pH/EC is predicted using only past pH/EC measurements, forecast climate, and planned irrigation/fertigation.
- Future pH/EC samples are withheld from training until they become available, preventing data leakage.

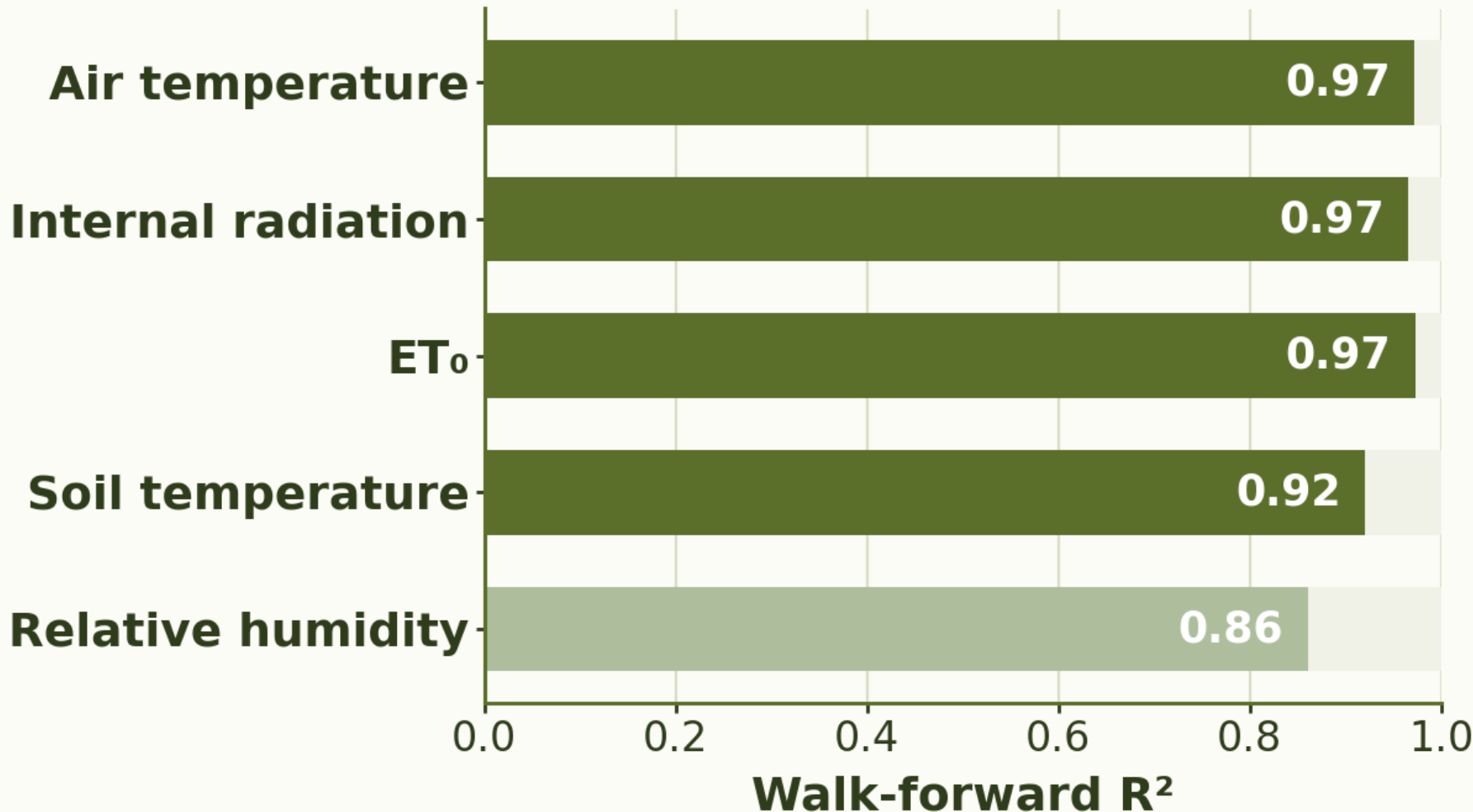
Walk-forward validation



07 Microclimate Model

- Predicts internal greenhouse microclimate from external weather and radiation.
- Generates continuous inputs for the root-zone model: temperature, RH, radiation, ET₀, and soil temperature.

Stage 1 — Microclimate model accuracy



08 Root-zone Model Results — Walk-forward Test

0.33 pH MAE

0.93 pH R²

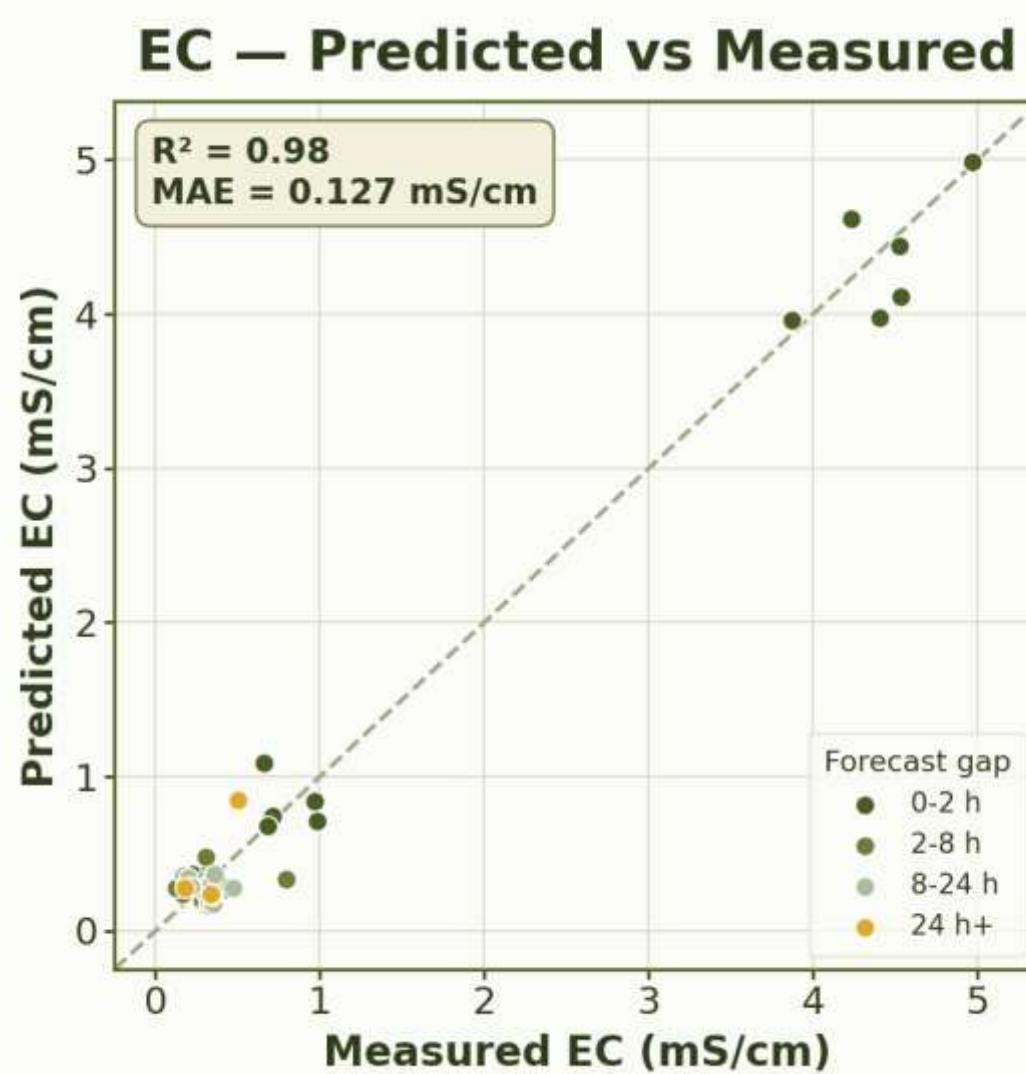
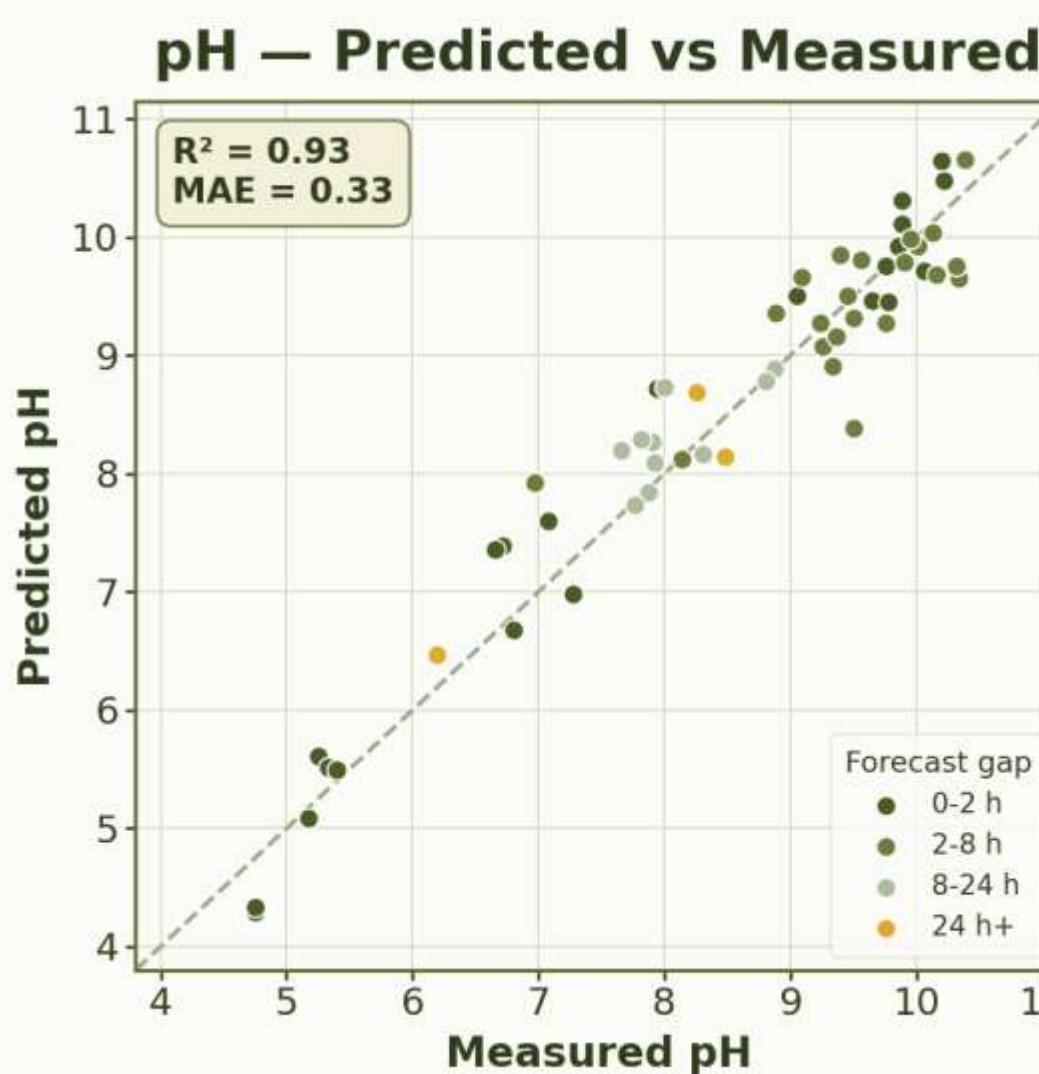
+62% pH MAE reduction

0.13 EC MAE

0.98 EC R²

+35% EC MAE reduction

n = 57 prediction intervals · MAE reduction vs. persistence baseline



09 Conclusions

- Prediction value** → Reduced MAE vs. baseline (+62% pH, +35% EC), supporting earlier pH/EC drift detection.
- Decision-support prototype** → Allows growers to test planned irrigation/fertigation actions before applying them.
- Soft-sensor feasibility** → Sparse samples can produce useful forecasts when combined with dense greenhouse data.

10 Challenges & Limitations

- Sparse pH/EC labels** → modeled Δ pH/ Δ EC between measured samples
- Missing climate inputs** → filled greenhouse gaps with microclimate predictions
- Irregular sampling** → used time-gap and window aggregation features
- Small EC changes** → modeled relative EC change and evaluated with MAE
- Limited generalization** → requires testing across more seasons, crops, and greenhouses